EXPERIMENT 15

Career Pathways and Future of Low-Code Development

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Subject: Low-Code Application Development using Zoho Creator

Project: EcoCycle – E-Waste Collection & Recycling Management

Aim

To study the career pathways in low-code development, to identify the skills required for different roles, and to understand the future growth and evolution of low-code platforms such as Zoho Creator, with reference to the EcoCycle project.

Introduction to Low-Code Development

Low-code development is an approach to building software mainly through visual tools and configuration instead of large amounts of traditional program code. Applications are assembled in visual development environments from predefined components and templates: forms for capturing data, reports and dashboards for viewing it, and workflows that define how information moves through a process. Zoho Creator is an example of such a platform, in which applications are built from forms, reports and workflows and extended with Deluge scripting.

In EcoCycle, a form captures each e-waste item, a workflow calculates its priority and moves it from Pending and Not Processed onwards, and reports and a dashboard show the staff which items and requests need attention.

Automation is a central feature of low-code platforms. Rules and triggers perform repetitive tasks without manual effort, such as calculating a priority score or e-mailing an alert for high-priority e-waste. Because much of the routine programming is handled by the platform, development effort is reduced and applications can be delivered faster. Low-code is also accessible to a wider group of people: professional developers use it to speed up their work, and business users with limited programming experience can take part in building simple applications.

However, low-code does not remove the need for technical understanding. Concepts such as data models, logic, integration and security remain important, and developers who understand them can design applications, extend them where the platform falls short and resolve problems when they occur.

Career Pathways in Low-Code Development

A range of professional roles is associated with low-code development. They differ in the depth of technical knowledge and the scope of responsibility. The roles are described below with an example of how each could relate to EcoCycle.

Role	Focus	Typical responsibilities	Example in this project
Low-Code Application Developer	Building applications on low-code platforms	Creating forms, reports and workflows; configuring application logic and automation	Builds the E-Waste Entry, Collection Request and Recycling Tracking forms and the four reports

Role	Focus	Typical responsibilities	Example in this project
Business Application Developer	Applications for specific business processes	Converting business requirements into working applications	Turns the collection-to-recycling process into the application
Automation Developer	Process automation	Designing automated workflows; reducing manual work with triggers	Writes the Deluge workflows for priority, statuses, date check and alert e-mail
Integration Developer	Connecting systems	Working with APIs and external services; data exchange between platforms	Connects Zoho CRM, builds the Custom API and plans the REST API, Maps and SMS integrations
Low-Code Consultant	Advising on low-code	Understanding requirements; recommending suitable low-code solutions	Advises an institution on whether e-waste tracking fits a low-code platform
Solution Architect	Designing complete solutions	Defining architecture across low-code, pro-code and enterprise systems	Designs the hybrid architecture of Experiment 13
Pro-Code Developer / Hybrid Developer	Extending platforms with code	Writing custom functions, APIs and services where low-code falls short	Builds custom functions or Zoho Catalyst services beyond the platform's limits

Skills Required

Skill area	What it covers	Practised in EcoCycle
Platform knowledge	Forms, reports, pages, workflows, connections	Three forms, four reports, dashboard, workflows
Scripting	Deluge and basic programming logic	Priority calculation, validation and alert scripts
Data modelling	Entities, relationships, lookups, validation	Data dictionary; lookups recommended between forms
Integration	APIs, connections, OAuth2	Zoho CRM connection; Custom API secured with OAuth2
AI tools	Using and checking AI features	Zia chat agent with read-only access
Security	Roles, permissions, data protection	Read-only agent; roles proposed in Experiment 8
Testing and documentation	Test cases, defect logs, user manuals	Chapter 6 test cases and defect log; user manual
Business understanding	Translating a process into an application	Hazard, age and condition factors of e-waste

Career Progression

Stage	Typical role	Focus
1	Low-Code Application Developer	Forms, reports and basic workflows (for example, EcoCycle's forms and reports)
2	Automation / Business Application Developer	Deluge workflows and process automation

Stage	Typical role	Focus
3	Integration Developer	Connections, APIs and external services
4	Low-Code Consultant / Hybrid Developer	Choosing approaches and adding pro-code extensions
5	Solution Architect	Complete architectures across low-code, pro-code and enterprise systems

At each stage technical depth and responsibility increase, and skills in databases, APIs, programming, automation and business understanding become more important.

Career Opportunities

Low-code professionals can work in organisations and areas such as the following:

- Enterprise application development
- Business process automation
- Digital transformation
- IT consulting
- Workflow automation
- System integration
- Cloud application development
- Internal business tools
- Data-driven applications

Because low-code platforms are used to automate and digitise business processes, career opportunities can exist across different industries, including environmental management, where applications such as EcoCycle help institutions record e-waste and follow it to recycling.

Future Career Outlook

Low-code development is likely to evolve rather than simply replace conventional software development. Several areas are expected to grow in importance:

- AI-assisted development, such as generating an application from a prompt and chat agents like the EcoCycle Data Assistant
- Hybrid low-code and pro-code development, as in the three-tier architecture of Experiment 13
- Integration skills (APIs, connections, the planned REST API of EcoCycle)
- Cloud technologies, data management and security
- Business understanding and solution architecture

As platforms and AI tools take over more repetitive implementation, future developers may spend less time on routine work and more time on problem solving, architecture, customisation, integration and business outcomes.

Result

The career pathways in low-code development were studied, together with the skills each role needs and the way each role could relate to the EcoCycle project. The future outlook of low-code platforms such as Zoho Creator was reviewed.

Conclusion

Low-code development provides an accessible and efficient approach to application development, and it is creating a growing range of career opportunities. Visual tools, reusable components and automation allow applications to be delivered faster and allow both professional developers and business users to contribute. A career can begin with low-code application development, such as building the forms and workflows of EcoCycle, and advance towards automation, integration, consulting, solution development and architecture.

The future of low-code development will be strongly influenced by AI, automation, APIs, cloud technologies and collaboration between low-code and pro-code approaches. Developer roles are expected to evolve rather than disappear. A strong future-ready developer should therefore combine low-code platform knowledge with programming fundamentals, databases, APIs, AI tools and business understanding.